

Same synergy. Different added information.

Finite categorical MGW shared exclusions · signed population atoms · nats

Shared law: A, B and U are mutually independent fair bits.

Baseline A; target Y = (A, B); added candidate C is U or B.

Both source pairs have $P(A = a, C = c) = 1/4$ for a, c in $\{0, 1\}$.

Signed MGW atom	Add C = U	Add C = B
	Unrelated to target Y	Missing target bit
Redundancy	$\ln(4/3)$	$\ln(4/3)$
Unique baseline A	$\ln(3/2)$	$\ln(3/2)$
Unique added C	$-\ln(4/3)$	$\ln(3/2)$
Net synergy S	$\ln(4/3)$	$\ln(4/3)$
Added information $I(C; Y \mid A)$	0	$\ln 2$

Unique added C + net synergy = added conditional MI

U: $-\ln(4/3) + \ln(4/3) = 0$

B: $\ln(3/2) + \ln(4/3) = \ln 2$

Synergy ties; the full signed decomposition retains the CMI difference.

Signed cancellation is published in MGW (RndErr, §VI.C / Table III).

Matched comparison: priority unestablished; no estimator or deployment claim.